
Relative Wasserstein Spatial Depth for Cluster Number Selection in Distributional Data

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Abstract

Modern datasets are increasingly represented as probability distributions, motivating clustering methods that operate in Wasserstein space. While recent advances extend classical clustering to distribution-valued data, most existing approaches assume that the number of clusters is known a priori, an assumption that rarely holds in practice. In this work, we address the problem of cluster number selection and robust clustering in Wasserstein space. We propose a new criterion, Relative Wasserstein Spatial Depth (RWSD), which leverages Wasserstein geometry and depth-based centrality to provide a robust and interpretable measure of cluster separation tailored to distributional data. In addition, we develop a Wasserstein K-median clustering algorithm that improves robustness to heavy-tailed variability and outliers. Empirical results demonstrate that the proposed method consistently outperforms classical distance-based criteria, including the silhouette score and Davies–Bouldin index, particularly in challenging settings with outliers, overlapping clusters, and heterogeneous cluster structures.

1 Introduction

Contemporary data are increasingly complex and multi-faceted. In many applications, each data object is more naturally represented as a probability distribution rather than a vector in Euclidean space, such as distributions of gene expression levels across spatial locations, connectivity profiles across brain regions, high-resolution images, and media streams. Such representation captures the intrinsic variability and geometric structure of the data, while providing a low-dimensional embedding in the Wasserstein space [Panaretos and Zemel, 2019, Bachoc et al., 2024].

As an important exploratory analysis technique, clustering generally groups objects based on their similarity. When data objects are distributions, classical clustering methods based on Euclidean geometry may fail to capture meaningful differences in shape, dispersion, or support. Clustering techniques adapting to Wasserstein space are in high demand, which respects the underlying geometry of probability distributions.

Related Work Several efforts have been made to devise clustering techniques for distribution-valued data. Trimmed K -barycenters method [Del Barrio et al., 2019] improves stability and robustness by trimming the most discrepant distributions. A modified Wasserstein distance proposed in [Verdinelli and Wasserman, 2019] is computationally more tractable and accelerates various clustering procedures. The work [Chazal et al., 2021] embeds distributions into a finite Euclidean space, enabling the application of classical clustering methods. The Wasserstein K -means clustering framework [Zhuang et al., 2022] generalizes classical K -means to Wasserstein space through a distance-based formulation

$$\min_{\{C_k\}_{k=1}^K} \left\{ \sum_{k=1}^K \frac{1}{|C_k|} \sum_{\mu_i, \mu_j \in C_k} \mathcal{W}_2^2(\mu_i, \mu_j), \cup_{k=1}^K C_k = [n] \right\},$$

and proposes a semidefinite program (SDP) relaxation to approximate the solution. Spectral clustering is also generalized to Wasserstein space by constructing an adjacency matrix of pairwise Wasserstein distances [Wang et al., 2024]. Together, these works open a new direction for unsupervised pattern discovery in Wasserstein space.

Motivation All these clustering methods assume that the number of clusters K is known *a priori*, which is not realistic in many applications. Selecting a suitable K remains a challenge in practice [Still and Bialek, 2004].

For clustering in Euclidean space, several measurements (statistics) have been proposed to pick K and assess how statistically meaningful a clustering is. *Silhouette Score* [Rousseeuw, 1987] is among the most widely used, which evaluates how similar each point is to its own cluster compared to other clusters. The Wasserstein version of silhouette score can be defined as follows. Given a particular clustering scheme $\{\mathcal{C}_1, \dots, \mathcal{C}_K\}$, a distribution μ_i assigned to $\mathcal{C}_{k_i}$ has silhouette score being the difference between its within-cluster dissimilarity and minimum out-of-cluster dissimilarity,

$$s_i = (b_i - a_i) / \max(a_i, b_i), \quad a_i = \frac{1}{|\mathcal{C}_{k_i}| - 1} \sum_{\mu_j \in \mathcal{C}_{k_i}} \mathcal{W}_2(\mu_i, \mu_j), \quad b_i = \min_{\ell \neq k_i} \frac{1}{|\mathcal{C}_\ell|} \sum_{\mu_j \in \mathcal{C}_\ell} \mathcal{W}_2(\mu_i, \mu_j). \quad (1)$$

A larger s_i indicates better separation of $\mathcal{C}_{k_i}$ from other clusters. In practice, one computes the average silhouette score $\bar{s}(K) = \frac{1}{n} \sum_{i=1}^n s_i(K)$ for different K 's, and selects $\hat{K} = \arg \max_K \bar{s}(K)$. The silhouette score is sensitive to overlapping or nonconvex clusters, as it relies on averages of pairwise distances [Liu et al., 2010]. Another widely used measurement is the *Davies–Bouldin Index (DBI)* [Davies and Bouldin, 1979], which evaluates the average of ratios between within-cluster scattering and between-cluster separation. The Wasserstein version of DBI is defined as

$$\text{DBI}(K) = \frac{1}{K} \sum_{k=1}^K \max_{\ell \neq k} \frac{R_k + R_\ell}{\mathcal{W}_2(\bar{\mu}_k, \bar{\mu}_\ell)}, \quad R_k = \frac{1}{|\mathcal{C}_k|} \sum_{\mu_i \in \mathcal{C}_k} \mathcal{W}_2(\mu_i, \bar{\mu}_k), \quad (2)$$

where R_k quantifies the within-cluster scattering of $\mathcal{C}_k$ and $\bar{\mu}_k$ is the representative of $\mathcal{C}_k$ (e.g. its Wasserstein barycenter). A smaller DBI indicates tighter and more separated clusters, and the best K is the one that minimizes DBI. DBI is sensitive to outliers and blurred cluster boundaries [Liu et al., 2010, Zaki and Meira, 2014], and it can be unstable when the cluster scale changes [Halkidi and Vazirgiannis, 2001]. In Euclidean space, Jörnsten et al. [2002] proposed the relative L_1 -depth for cluster validation.

Although these measurements can be extended to Wasserstein space by replacing Euclidean distances with $\mathcal{W}_2$, they suffer from limited robustness (being sensitive to outliers) and may neglect the rich geometry of Wasserstein space. These limitations motivate the development of a robust and interpretable measure tailored to Wasserstein space.

Our Contribution. We introduce a new criterion, *Relative Wasserstein spatial depth (RWSD)*, for selecting the number of clusters K when performing Wasserstein K -center clustering. We also develop a *Wasserstein K -median clustering* algorithm that is robust to outliers and heavy tailed dispersions.

2 Methodology

Throughout the paper, let $\mathcal{P}_2(\mathbb{R}^d)$ be the collection of probability distributions on $\mathbb{R}^d$ with finite second moment, among which the subset of absolutely continuous distributions is denoted as $\mathcal{P}_2^{a.c.}(\mathbb{R}^d)$. Let $\mathcal{P}(\mathcal{P}_2(\mathbb{R}^d))$ be the collection of distributions of distributions in $\mathcal{P}_2(\mathbb{R}^d)$.

2.1 Wasserstein spatial depth

Classical statistical depth can be viewed as a multivariate analogue of the classical quantiles or ranks for univariate data. It provides a robust measure of centrality and outlyingness for multivariate data or functional observations [Zuo and Serfling, 2000]. Unlike distance-based measures, depth functions are affine-invariant and resistant to outliers, making them particularly suitable for analyzing nonconvex or heterogeneous data structures. A new notion of depth defined on Wasserstein space is

recently proposed in [Bachoc et al., 2024], named as *Wasserstein spatial depth* (WSD). The WSD of a distribution $Q \in \mathcal{P}_2^{a.c.}(\mathbb{R}^d)$ with respect to a population of distributions $\mathbf{P} \in \mathcal{P}(\mathcal{P}_2(\mathbb{R}^d))$ is defined as

Definition 2.1 (Wasserstein spatial depth [Bachoc et al., 2024]).

$$\text{WSD}(Q; \mathbf{P}) := 1 - \left(\int \left\| \mathbb{E}_{P \sim \mathbf{P}} \left[\frac{x - T_{Q,P}(x)}{\mathcal{W}_2(Q, P)} \right] \right\|^2 dQ(x) \right)^{1/2}.$$

When $\mathbb{P}_{P \sim \mathbf{P}}\{\mathcal{W}_2(P, Q) = 0\} \neq 0$, we set $\frac{x - T_{Q,P}(x)}{\mathcal{W}_2(P, Q)} = 0$ by default.

It provides a way to rank and order distributions, enabling the development of rank-based clustering techniques and inferential tools for Wasserstein space. WSD is shown to preserve critical properties of conventional statistical depths, notably, ranging within $[0, 1]$, transformation and geodesic invariance, vanishing at infinity, reaching a maximum at the geometric median, and continuity. Additionally, the population WSD has a straightforward plug-in estimator based on sampled empirical distributions, which is consistent and asymptotically normal.

In practice, we only have access to a sample of empirical distributions $\hat{\mu}_{1,m}, \hat{\mu}_{2,m}, \dots, \hat{\mu}_{n,m}$: first draw $\mu_1, \dots, \mu_n \stackrel{iid}{\sim} \mathbf{P}$ and then draw data points $x_{i,1}, \dots, x_{i,m}$ for each μ_i . Therefore, the empirical version of WSD is computed as

$$\text{WSD}(\hat{\mu}_{i,m}; \hat{\mathbf{P}}_{n,m}) = 1 - \left(\frac{1}{m} \sum_{j=1}^m \left\| \frac{1}{n} \sum_{\ell \neq i} \frac{X_{i,j} - T_{\hat{\mu}_{i,m}, \hat{\mu}_{\ell,m}}(X_{i,j})}{\mathcal{W}_2(\hat{\mu}_{i,m}, \hat{\mu}_{\ell,m})} \right\|^2 \right)^{1/2}, \quad (3)$$

where $T_{\hat{\mu}_{i,m}, \hat{\mu}_{\ell,m}}$ is empirical optimal transport map, with implementation details deferred to Appendix A.3.

2.2 Relative Wasserstein spatial depth

Building on the concept of WSD, we propose *Relative Wasserstein Spatial Depth* (RWSD) as a robust index for selecting an optimal number of cluster K . The RWSD of a particular distribution Q is defined as

$$\text{RWSD}(Q) = \text{WSD}(Q; \mathcal{C}_{k_Q}) - \max_{j \neq k_Q} \text{WSD}(Q; \mathcal{C}_j), \quad (4)$$

where k_Q indicates which cluster Q belongs to. RWSD compares within- and between-cluster depths, emphasizing centrality and separation rather than distance or variance. A larger $\text{RWSD}(Q)$ indicates that Q is considerably deeper within its own cluster than in others, implying a confident assignment.

Let $Q_1, Q_2, \dots, Q_N$ be the set of distributions to be clustered. The optimal number of clusters K^* can be determined by maximizing the average RWSD

$$K^* = \arg \max_K \frac{1}{N} \sum_{i=1}^N \text{RWSD}_K(Q_i), \quad (5)$$

where $\text{RWSD}_K(Q_i)$ denotes the relative WSD of Q_i under the clustering solution with K clusters. This criterion favors partitions that are both internally cohesive and externally well separated in terms of Wasserstein geometry. Compared to classical distance-based indices (silhouette score and DBI), RWSD respects Wasserstein geometry and is invariant under common transformations (*e.g.*, scaling). Additionally, it is robust to largely dispersed, unbalanced, and overlapping clusters (shown in Section 4). The empirical version of RWSD is computed by simply plugging in the empirical WSD.

To implement RWSD, we run Algorithm 1.

Algorithm 1 Select Optimal Number of Clusters K by RWSD

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1: Input: a collection of probability  $\{\mu_1, \dots, \mu_n\} \subset \mathcal{P}_2(\mathbb{R}^d)$ , a candidate set  $\mathcal{K}$  containing possible
   numbers of clusters
2: for each  $K \in \mathcal{K}$  do
3:   Run Wasserstein  $K$ -centers to partition  $\{\mu_i\}_{i=1}^n$  into  $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_K$ .
4:   for  $i = 1$  to  $n$  do
5:      $\text{RWSD}_K(\mu_i) \leftarrow \text{WSD}(\mu_i; \mathcal{C}_{k(\mu_i)}) - \max_{j \neq k(\mu_i)} \text{WSD}(\mu_i; \mathcal{C}_j)$ 
6:   end
7:    $\overline{\text{RWSD}}(K) \leftarrow \frac{1}{n} \sum_{i=1}^n \text{RWSD}_K(\mu_i)$ 
8: end
9:  $K^* \leftarrow \arg \max_{K \in \mathcal{K}} \overline{\text{RWSD}}(K)$ 
10: Output: the optimal cluster count  $K^*$ 
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In the above procedure, the “Wasserstein K -centers step” can be either Wasserstein K -means or Wasserstein K -medians. The details are deferred to Appendix A. The overall complexity of Algorithm 1 for a single K is dominated by the pairwise OT computations, with $\mathcal{O}(m^2/\varepsilon^2)$ per OT computation (regularized OT); a detailed breakdown is provided in Appendix E.1.

3 Theoretical guarantee

The empirical RWSD_K is a function of the empirical K -partition and the empirical WSD. Hence, the deviation of empirical RWSD from its population version comes from the randomness in the empirical Wasserstein K -means clustering and empirical WSD. Establishing the consistency of empirical RWSD therefore requires two ingredients: (a) consistency of the empirical WSD, which is established Bachoc et al. [2024, Theorem 6.3]; (b) consistency of Wasserstein K -means clustering, *i.e.*, convergence of the empirical cluster centers to the well-defined population analogies as $n \rightarrow \infty$, a strong consistency in the sense of Pollard [1981].

Theorem 3.1 establishes the consistency of Wasserstein K -means. Theorem 3.2 shows that the empirical RWSD is a consistent estimator of its population counterpart if the clustering is consistent.

Let (X, d) be a Polish metric space and let $\mathcal{P}_2(X)$ denote the space of Borel probability measures on X with finite second moment, endowed with the 2-Wasserstein distance. We observe i.i.d. random objects taking values in $\mathcal{P}_2(X)$

$$\mu_1, \mu_2, \dots, \mu_n \sim \mathbf{P}, \quad \mathbf{P} \in \mathcal{P}(\mathcal{P}_2(X))$$

with the empirical of $\mathbf{P}$ denoted as $\mathbf{P}_n := \frac{1}{n} \sum_{i=1}^n \delta_{\mu_i}$.

Given a finite set $A \subset \mathcal{P}_2(X)$ and a distribution of distributions $\mathbf{P} \in \mathcal{P}(\mathcal{P}_2(X))$, consider the function defined in [Pollard, 1981]

$$\Phi(A, \mathbf{P}) := \int \min_{\nu \in A} \mathcal{W}_2^2(\mu, \nu) \mathbf{P}(d\mu). \quad (6)$$

The minimizer and minimal value of $\Phi(A, \mathbf{P})$ over all A ’s with cardinality bounded by K are, respectively,

$$A_K^* \in \arg \min_{|A| \leq K} \Phi(A, \mathbf{P}), \quad m_K(\mathbf{P}) := \inf \{ \Phi(A, \mathbf{P}) : |A| \leq K \}.$$

Their empirical versions are

$$A_{n,K}^* \in \arg \min_{|A| \leq K} \Phi(A, \mathbf{P}_n), \quad m_K(\mathbf{P}_n) := \inf \{ \Phi(A, \mathbf{P}_n) : |A| \leq K \}.$$

Under compactness of the base space (X, d) , the Wasserstein space $\mathcal{P}_2(X)$ is itself compact under $\mathcal{W}_2$; hence both population and empirical minimizers exist.

Theorem 3.1 (Consistency of Wasserstein k -means). *Assume:*

- (i) (X, d) is a compact metric space. Hence $\mathcal{P}_2(X)$ is compact under the $\mathcal{W}_\infty$ metric.

(ii) For some (and therefore every) $\mu_0 \in \mathcal{P}_2(X)$,

$$\int_{\mathcal{P}_2(X)} \mathcal{W}_2(\mu, \mu_0)^2 \mathbf{P}(d\mu) < \infty.$$

(iii) For each $k = 1, \dots, K$, the population minimizer A_k^* satisfying $\Phi(A_k^*, \mathbf{P}) = m_k(\mathbf{P})$ is unique up to permutation.

(iv) The population k -means values are strictly separated across consecutive model sizes:

$$m_{k-1}(\mathbf{P}) > m_k(\mathbf{P}), \quad k = 2, \dots, K.$$

Then the empirical k -means centers are strongly consistent:

$$A_{n,k}^* \longrightarrow A_k^* \quad \text{a.s. in the Hausdorff metric,}$$

and the empirical optimal values satisfy

$$\Phi(A_{n,k}^*, \mathbf{P}_n) \rightarrow m_k(\mathbf{P}) \quad \text{a.s.}$$

We next show that whenever the Wasserstein K -center clustering method produces strongly consistent cluster centers, the corresponding empirical RWSD is a consistent estimator of its population counterpart.

Theorem 3.2 (Consistency of empirical RWSD under consistent clustering). *Let $\mathbf{P} \in \mathcal{P}(\mathcal{P}_2(\mathbb{R}^d))$ be atomless, and suppose the Wasserstein K -center clustering method produces empirical cluster centers $\{\hat{A}_{n,k}\}_{k=1}^K$ that converge almost surely in the Hausdorff metric induced by $\mathcal{W}_2$ to a set of unique population centers $\{A_k^*\}_{k=1}^K$. Assume further that the population partition $\{\mathcal{C}_k\}_{k=1}^K$ induced by $\{A_k^*\}_{k=1}^K$ is well-separated, in the sense that $\mathbf{P}$ -almost every μ has a unique nearest center. Then the empirical RWSD satisfies*

$$\text{RWSD}_K(\hat{\mu}_{i,m}) \xrightarrow{P} \text{RWSD}_K(\mu_i) \quad \text{as } n, m \rightarrow \infty,$$

for $\mathbf{P}$ -almost every μ_i .

Combining Theorem 3.1 and Theorem 3.2 immediately yields the consistency of empirical RWSD induced by Wasserstein K -means clustering.

Corollary 1 (Consistency of empirical RWSD under Wasserstein K -means). *Under the conditions of Theorem 3.1 and Theorem 3.2, the empirical RWSD satisfies*

$$\text{RWSD}_K(\hat{\mu}_{i,m}) \xrightarrow{P} \text{RWSD}_K(\mu_i) \quad \text{as } n, m \rightarrow \infty,$$

for $\mathbf{P}$ -almost every μ_i .

The proofs for the above theorems and corollary are given in Appendix B.2.

4 Numerical simulations

Given the sample set of empirical distributions, we implement Wasserstein K -center clustering with three selection criteria, the silhouette score, the DBI, and RWSD, and record their selected optimal K values. We choose silhouette score and Davies and Bouldin index for comparison because these two criteria are the most widely used and computationally efficient; additional discussion on other criteria is provided in Appendix D. In all the simulations, we use a two-stage sampling scheme:

- Step 1: draw distributions $\mu_1^k, \mu_2^k, \dots, \mu_{n_k}^k$ from the k -th Cluster;
- Step 2: draw m data points $\mathbf{x}_1^{i,k}, \mathbf{x}_2^{i,k}, \dots, \mathbf{x}_m^{i,k}$ from each μ_i^k , $i = 1, \dots, n_k$, $k = 1, \dots, K$.

Throughout this section, we choose $n_k = 200$, $m = 50$ for all $k = 1, \dots, K$, and each reported quantity in the tables below represents the average of 20 independent replications with $\pm$ one standard deviation.

4.1 Clusters with imbalanced dispersion

Consider two highly heterogeneous clusters of distributions,

$$\begin{aligned}\text{Cluster 1} &= \left\{ N(\mathbf{c}_1, \mathbf{I}_{2 \times 2}), \mathbf{c}_1 \sim \text{Unif}[\mathcal{B}_{100}(\mathbf{0})] \right\}, \\ \text{Cluster 2} &= \left\{ N(\mathbf{c}_2, \mathbf{I}_{2 \times 2}), \mathbf{c}_2 \sim \text{Unif}[\mathcal{B}_{0.1}((120, 0)^\top)] \right\},\end{aligned}$$

where $\mathcal{B}_{0.1}((120, 0)^\top)$ represents a ball centered at $(120, 0)^\top$ with radius 0.1. As shown in Figure 1, the distributions in Cluster 1 (blue) are substantially more dispersed than those in Cluster 2 (orange), and Cluster 2 is almost part of Cluster 1, which creates extremely difficulty in separating two clusters.

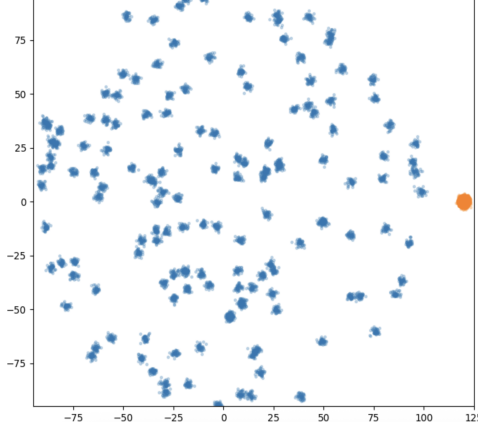


Figure 1: Two clusters of unbalanced dispersions.

Table 1: The optimal K^* chosen by three criteria.

K	Silhouette	RWSD	DBI
2	.656 ($\pm .012$)	.358 ($\pm .005$)	.602 ($\pm .031$)
3	.654 ($\pm .010$)	.354 ($\pm .014$)	.814 ($\pm .049$)
4	.669 ($\pm .010$)	.351 ($\pm .017$)	.727 ($\pm .033$)
5	.664 ($\pm .010$)	.342 ($\pm .018$)	.761 ($\pm .065$)
6	.667 ($\pm .009$)	.343 ($\pm .015$)	.768 ($\pm .033$)
7	.671 ($\pm .008$)	.337 ($\pm .013$)	.757 ($\pm .038$)
8	.672 ($\pm .010$)	.337 ($\pm .010$)	.767 ($\pm .053$)
9	.674 ($\pm .010$)	.336 ($\pm .010$)	.763 ($\pm .059$)
10	.676 ($\pm .011$)	.336 ($\pm .012$)	.762 ($\pm .043$)
K^*	10	2	2

The optimal number of clusters under different selection criteria, including RWSD (Algorithm 1), silhouette score, and DBI are summarized in Table 1. Distance-based selection indices show a tendency toward larger values of K because they rely on distance-based dispersions. By contrast, RWSD measures each distribution's relative centrality within its assigned cluster, and remain stable under extremely heterogeneous clusters.

4.2 Clusters with unbalanced separations

Consider three clusters with unbalanced separations,

$$\text{Cluster } k = \left\{ N(\mathbf{c}_k, \sigma^2 \mathbf{I}_{2 \times 2}), \mathbf{c}_k \sim \text{Unif}[\mathcal{B}_k] \right\}, \quad k = 1, 2, 3,$$

where $\mathcal{B}_1 = [-1.8, -1.7] \times [0, 0.5]$, $\mathcal{B}_2 = [-0.3, -0.2] \times [0, 0.5]$, $\mathcal{B}_3 = [20, 21] \times [0, 0.5]$, $\sigma = 0.12$. As shown in Figure 2, Clusters 1 and 2 lie closer together while Cluster 3 is isolated. Silhouette and

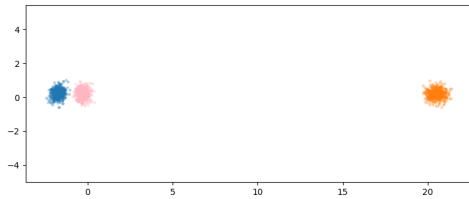


Figure 2: Clusters with unbalanced separations.

Table 2: The optimal K^* chosen by three criteria.

K	Silhouette	RWSD	DBI
2	.967 ($\pm .001$)	.398 ($\pm .001$)	.049 ($\pm .001$)
3	.913 ($\pm .002$)	.402 ($\pm .001$)	.127 ($\pm .004$)
4	.764 ($\pm .029$)	.375 ($\pm .003$)	.438 ($\pm .065$)
5	.673 ($\pm .040$)	.355 ($\pm .006$)	.617 ($\pm .043$)
6	.589 ($\pm .029$)	.334 ($\pm .005$)	.709 ($\pm .038$)
K^*	2	3	2

DBI both select $K = 2$, merging the two nearby clusters because the distant Cluster 3 dominates all pairwise distances. RWSD correctly identifies $K = 3$, as depth-based centrality remains sensitive to local structure.

4.3 Clusters with heterogeneous scales

Consider three clusters of centered Gaussian distributions with very different variances,

$$\text{Cluster } k = \left\{ N(\mathbf{0}, \sigma_k^2 \mathbf{I}_{2 \times 2}), \sigma_k \sim \text{Triangular}[a_k, b_k, c_k] \right\}, \quad k = 1, 2, 3,$$

where $(a_1, b_1, c_1) = (0.0, 0.2, 0.4)$, $(a_2, b_2, c_2) = (1.0, 1.2, 1.4)$, $(a_3, b_3, c_3) = (5.0, 5.2, 5.4)$. The data clouds from the three clusters are marked as dark green, green, and light green in Figure 3. Silhouette selects $K = 2$ because the smallest-variance group forms a compact core dominating

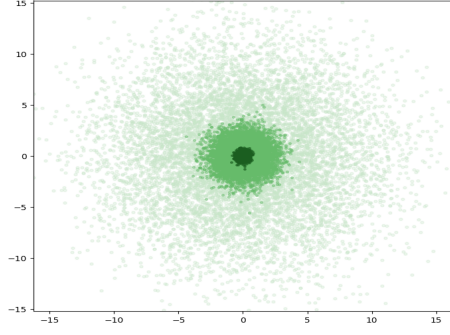


Figure 3: Clusters with heterogeneous scales.

Table 3: The optimal K^* chosen by three criteria.

K	Silhouette	RWSD	DBI
2	.660 ($\pm .003$)	.360 ($\pm .002$)	.569 ($\pm .005$)
3	.511 ($\pm .003$)	.365 ($\pm .002$)	.553 ($\pm .004$)
4	.310 ($\pm .028$)	.186 ($\pm .045$)	1.732 ($\pm .114$)
5	.219 ($\pm .077$)	.113 ($\pm .048$)	2.153 ($\pm .219$)
6	.173 ($\pm .066$)	.082 ($\pm .042$)	2.345 ($\pm .183$)
7	.171 ($\pm .065$)	.079 ($\pm .042$)	2.408 ($\pm .148$)
8	.160 ($\pm .057$)	.071 ($\pm .037$)	2.421 ($\pm .138$)
9	.154 ($\pm .045$)	.070 ($\pm .035$)	2.381 ($\pm .152$)
10	.152 ($\pm .045$)	.068 ($\pm .034$)	2.393 ($\pm .134$)
K^*	2	3	3

dispersion comparisons. DBI improves slightly but still struggles under heterogeneous within-cluster spread. RWSD recovers $K = 3$, as Wasserstein depth reflects how each cloud's dispersion positions it relative to the cluster center.

4.4 Clusters contaminated with outliers

Consider three clusters $\text{Cluster } k = \left\{ N(\mathbf{c}_k, 0.08^2 \mathbf{I}_{2 \times 2}), \mathbf{c}_k \sim N(\mathbf{a}_k, 0.6^2 \mathbf{I}_{2 \times 2}) \right\}$ for $k = 1, 2, 3$ with $\mathbf{a}_1 = (-0.8, 0)$, $\mathbf{a}_2 = (0.8, 0)$, $\mathbf{a}_3 = (0, 0.8\sqrt{3})$. Every cluster is contaminated with 40% outliers, where each outlier distribution is *iid* drawn from

$$\left\{ N(\mathbf{c}, 0.08^2 \mathbf{I}_{2 \times 2}), \mathbf{c} \sim \text{Unif}[\mathcal{R}] \right\},$$

where $\mathcal{R} = [-1.6, 1.6] \times [-0.8, 0.8 + 0.8\sqrt{3}]$ is a rectangle.

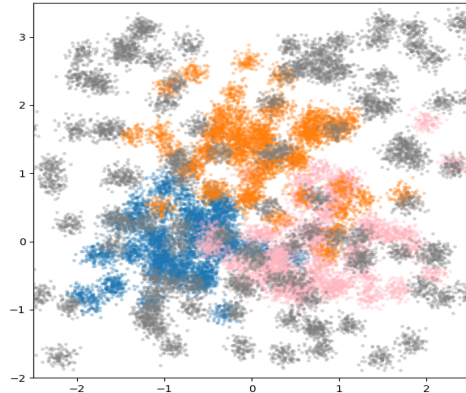


Figure 4: Clusters (colored) contaminated with outliers (gray).

Table 4: The optimal K^* chosen by three criteria.

K	Silhouette	RWSD	DBI
2	.318 ($\pm .018$)	.284 ($\pm .007$)	1.36 ($\pm .082$)
3	.357 ($\pm .018$)	.308 ($\pm .009$)	1.03 ($\pm .048$)
4	.328 ($\pm .030$)	.297 ($\pm .006$)	1.06 ($\pm .085$)
5	.331 ($\pm .023$)	.295 ($\pm .010$)	.997 ($\pm .086$)
6	.328 ($\pm .022$)	.292 ($\pm .008$)	.962 ($\pm .058$)
7	.328 ($\pm .020$)	.290 ($\pm .008$)	.942 ($\pm .059$)
8	.326 ($\pm .025$)	.289 ($\pm .009$)	.914 ($\pm .064$)
9	.328 ($\pm .018$)	.288 ($\pm .006$)	.903 ($\pm .056$)
10	.330 ($\pm .020$)	.287 ($\pm .007$)	.893 ($\pm .053$)
K^*	3	3	10

Shown in the above results, distance-based indices are largely affected by the outliers and split the true components into smaller artificial subclusters, while depth-based RWSD is robust to overlapping and outliers.

4.5 Clusters with heavy-tailed dispersion

Consider three clusters, for $k = 1, 2, 3$,

$$\text{Cluster } k = \left\{ N(\mathbf{c}_k, 0.2^2 \mathbf{I}_{2 \times 2}), \mathbf{c}_k = \mathbf{a}_k + (r \cos \theta, r \sin \theta) \right\},$$

where $\mathbf{a}_1 = (-1, 0)$, $\mathbf{a}_2 = (1, 0)$, $\mathbf{a}_3 = (0, \sqrt{3})$, $r \sim \text{Pareto}(r_{\min} = 0.35, \alpha = 0.45)$ truncated at $r_{\max} = 5.0$, and $\theta \sim \text{Unif}[0, 2\pi)$. Each cluster has a core and heavy-tailed peripheral data clouds.

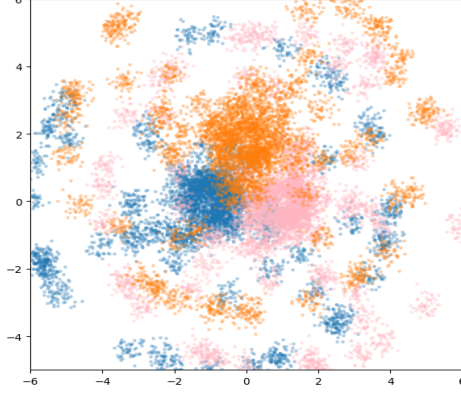


Figure 5: Clusters with heavy-tailed dispersion.

Table 5: The optimal K^* chosen by three criteria.

K	Silhouette	RWSD	DBI
2	.287 ($\pm .011$)	.278 ($\pm .004$)	1.667 ($\pm .037$)
3	.322 ($\pm .005$)	.307 ($\pm .002$)	1.282 ($\pm .029$)
4	.296 ($\pm .021$)	.289 ($\pm .011$)	1.178 ($\pm .030$)
5	.306 ($\pm .016$)	.284 ($\pm .006$)	1.037 ($\pm .039$)
6	.331 ($\pm .005$)	.284 ($\pm .005$)	.941 ($\pm .027$)
7	.369 ($\pm .006$)	.289 ($\pm .006$)	.839 ($\pm .007$)
8	.376 ($\pm .035$)	.293 ($\pm .009$)	.822 ($\pm .095$)
9	.373 ($\pm .030$)	.295 ($\pm .007$)	.837 ($\pm .058$)
10	.367 ($\pm .025$)	.293 ($\pm .009$)	.850 ($\pm .033$)
K^*	8	3	8

Silhouette and DBI both split the true components into smaller subclusters because heavy-tailed peripheral clouds inflate within-cluster dispersions. RWSD picks the correct K , thanks to its robustness to heavy tails.

Among the above settings, the first three achieve similar performance by Wasserstein K -means and K -medians, while Wasserstein K -medians beats K -means in 4.4 and 4.5. Regardless of Wasserstein K -means or K -medians, RWSD is uniformly better than Silhouette score and DBI.

5 Real data analysis

We apply RWSD to two real datasets, the MNIST handwritten digit dataset and a flow cytometry dataset from an acute myeloid leukemia cohort, to showcase its practical utility. Extra details are deferred to Appendix C.

5.1 MNIST digits

Each grayscale MNIST handwritten digit image of size 28×28 is treated as a discrete distribution over pixel coordinates with intensities normalized to sum to one, $\mu_i \in \mathcal{P}_2([0, 1]^2)$. We evaluate RWSD under different ground-truth numbers of clusters. The first setting (true $K = 3$) includes digits labeled as “1”, “2”, and “9”, with 30 images for each digit to form a total of $N = 90$ normalized empirical distributions. The second setting (true $K = 5$) includes digits “1”, “2”, “5”, “8”, and “9”, with 30 images for each digit to form $N = 150$ empirical distributions. For each $K \in \{2, \dots, 9\}$, we computed the silhouette score, RWSD, and DBI to pick the optimal K .

5.2 Acute myeloid leukemia (AML) flow cytometry cohort

We also apply RWSD to a flow cytometry dataset, BLAST110 Mocking et al., 2024, which collects bone marrow samples from 20 healthy individuals and 30 acute myeloid leukemia (AML) patients. Flow cytometry quantifies fluorescence-labeled protein markers at the single-cell level and is a central tool for characterizing hematopoietic sub-populations and leukemic phenotypes [Saeys et al., 2016]. The goal is to identify AML patients based on their abnormal distributions of protein markers on hematopoietic cells in bone marrow. The healthy population and AML patients are treated as two clusters, the flow cytometry profile of $d = 8$ biomarkers of each individual is a distribution (on

K	true $K = 3$			true $K = 5$		
	Silhouette	RWSD	DBI	Silhouette	RWSD	DBI
2	0.2429	0.1453	1.8272	0.1990	0.1395	1.8927
3	0.2794	0.1730	1.5550	0.2050	0.1344	1.7009
4	0.2754	0.1712	1.4019	0.2097	0.1397	1.6903
5	0.2541	0.1533	1.4831	0.2164	0.1398	1.7157
6	0.2872	0.1576	1.4330	0.1904	0.1204	1.9217
7	0.2601	0.1450	1.4536	0.1867	0.1176	1.9033
8	0.2447	0.1428	1.4703	0.1723	0.1173	1.7966
9	0.2176	0.1247	1.6573	0.1855	0.1073	1.9403
Best K	6	3	4	5	5	4

$\mathbb{R}^8$) from either cluster, and the $m = 1000$ cells (profiles) from each individual are $m = 1000$ data points drawn from the corresponding distribution. Details of cytometry preprocessing are deferred to Appendix C.2. As shown in Table 6, both silhouette score and RWSD work well on this dataset as the two populations (healthy vs. AML patients) have quite separated flow cytometry profiles, consistent with the existing reports [Mocking et al., 2024].

Table 6: RWSD and silhouette score pick the correct K .

K	Silhouette	RWSD	DBI
2	0.2156	0.1604	1.2391
3	0.2074	0.1435	1.2483
4	0.1450	0.0812	1.1107
5	0.1373	0.0604	0.9324
6	0.1549	0.0586	0.8751
Best K	2	2	6

6 Discussion and future work

We propose a new criterion, *Relative Wasserstein Spatial Depth* (RWSD), to select the optimal number of clusters in Wasserstein K -center clustering. RWSD favors partitions that are both internally cohesive and externally well separated in terms of centrality and Wasserstein geometry. Demonstrated by extensive simulations and real data analysis, RWSD is more robust than distance-based criteria under outliers, heavy dispersions, overlapped clusters, and unbalanced clusters. This advantage is largely attributed to its awareness of the nonlinear Wasserstein geometry.

On the theoretical side, we established a Pollard-type strong consistency result for Wasserstein K -means, which implies the consistency of RWSD. On the computational aspect, we devise Wasserstein K -medians clustering, which is robust to “irregular” clustering structure. Computational efficiency is largely improved by restricting the support size of Wasserstein barycenters, entropic regularized OT, and iteratively reweighted least squares. Observed in extensive simulations, the accuracy of RWSD depends on sample sizes but not very sensitive to dimension d —larger sample size leads to better performance; moderately large d does little worse.

Limitations We focus on scenarios where the clusters are geodesically convex and the WSD correctly induces a center–outward ordering. In such benign cases, RWSD effectively distinguishes within- and between-cluster geometry. It remains open to explore scenarios where the clusters are not geodesically convex, and global depth may mask valleys in the Wasserstein landscape leading to cluster merging. We emphasize using the proposed method alongside domain expertise, validation, and careful interpretation. A potential remedy would be *local* Wasserstein spatial depth, where the centrality of a distribution is evaluated within its small neighborhood.

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A Wasserstein K-center clustering

We describe two Wasserstein K -center clustering methods used in this work, namely Wasserstein K -means and Wasserstein K -medians. These two methods differ in choices of the centers of the clusters: the former uses Wasserstein barycenters, whereas the latter uses Wasserstein Fréchet medians.

A.1 Wasserstein K-means clustering

Given a collection of probability measures $\{\mu_i\}_{i=1}^n \subset \mathcal{P}_2(\mathbb{R}^d)$, Wasserstein K -means seeks a K -group partition by minimizing the within-cluster variation measured in the 2-Wasserstein metric:

$$\min_{\{\mathcal{C}_k, \nu_k\}_{k=1}^K} \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \mathcal{W}_2^2(\mu_i, \nu_k), \quad (7)$$

where $\mathcal{C}_1, \dots, \mathcal{C}_K$ form a partition of $\{1, \dots, n\}$, and each $\nu_k \in \mathcal{P}_2(\mathbb{R}^d)$ is the Wasserstein barycenter of the k th cluster.

Starting from an initial set of centroids $\nu_1^{(0)}, \dots, \nu_K^{(0)}$, the algorithm alternates between an assignment step and a barycenter update step. At iteration t , each distribution μ_i is assigned to its nearest current centroid:

$$\mathcal{C}_k^{(t)} = \left\{ i \in \{1, \dots, n\} : \mathcal{W}_2(\mu_i, \nu_k^{(t)}) \leq \mathcal{W}_2(\mu_i, \nu_j^{(t)}), \forall j = 1, \dots, K \right\}, \quad (8)$$

and the centroid of each cluster is then updated as the corresponding Wasserstein barycenter:

$$\nu_k^{(t+1)} = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{|\mathcal{C}_k^{(t)}|} \sum_{i \in \mathcal{C}_k^{(t)}} \mathcal{W}_2^2(\mu_i, \nu), \quad k = 1, \dots, K. \quad (9)$$

The full procedure is summarized in Algorithm 2.

Algorithm 2 Barycenter-based Wasserstein K -means clustering ([Zhuang et al., 2022])

```

1: Input: a collection of probability  $\{\mu_1, \dots, \mu_n\} \subset \mathcal{P}_2(\mathbb{R}^d)$ , number of clusters  $K$ , tolerance  $\varepsilon$ 
2: Output: cluster labels  $\{l_i\}_{i=1}^n$  and Wasserstein centroids  $\{\nu_k\}_{k=1}^K$ 
3: Initialize  $\nu_1^{(0)}, \dots, \nu_K^{(0)} \in \mathcal{P}_2(\mathbb{R}^d)$ 
4:  $t \leftarrow 0$ 
5: repeat
6:   for  $i = 1, \dots, n$  do
7:      $l_i^{(t)} \leftarrow \arg \min_{1 \leq k \leq K} \mathcal{W}_2(\mu_i, \nu_k^{(t)})$ 
8:   end
9:   for  $k = 1, \dots, K$  do
10:     $\mathcal{C}_k^{(t)} \leftarrow \{i : l_i^{(t)} = k\}$ 
11:     $\nu_k^{(t+1)} \leftarrow \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{|\mathcal{C}_k^{(t)}|} \sum_{i \in \mathcal{C}_k^{(t)}} \mathcal{W}_2^2(\mu_i, \nu)$ 
12:   end
13:    $\Delta^{(t)} \leftarrow \max_{1 \leq k \leq K} \mathcal{W}_2(\nu_k^{(t+1)}, \nu_k^{(t)})$ 
14:    $t \leftarrow t + 1$ 
15: until  $\Delta^{(t-1)} \leq \varepsilon$ 
16: return  $\{l_i^{(t-1)}\}_{i=1}^n$  and  $\{\nu_k^{(t)}\}_{k=1}^K$ 

```

The algorithm can be viewed as an alternating minimization scheme in which the cluster labels are updated by nearest-barycenter assignment, and the barycenters are obtained by solving regularized semi-free support OT problems (detailed below).

The Wasserstein barycenter ν^* of multiple distributions $\{\mu_i\}_{i=1}^n \in \mathcal{P}_2(\mathbb{R}^d)$ is defined as

$$\nu^* = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{n} \sum_{i=1}^n \mathcal{W}_2^2(\nu, \mu_i). \quad (10)$$

In practice, the μ_i 's are empirical distributions and written as $\mu_i = \sum_{j=1}^{m_i} b_{ij} \delta_{y_{ij}}$. When the barycenter is allowed to have completely free support, the problem (10) admits a linear programming (LP) formulation over a polytope that couples the barycenter with each input distribution. Expressing the barycenter as $\nu = \sum_{p=1}^K a_p \delta_{x_p}$ with unknown supports $X = \{x_p\}_{p=1}^K$ and weights $\mathbf{a} \in \Delta^K$, one seeks transport plans $T^{(i)} \in \mathbb{R}_+^{K \times m_i}$ satisfying $T^{(i)} \mathbf{1} = \mathbf{a}$ and $(T^{(i)})^\top \mathbf{1} = \mathbf{b}_i$ and minimize the total cost $\sum_i \langle T^{(i)}, C(X, Y_i) \rangle$. A key structural fact is the sparsity of optimal barycenters: there exists an optimal discrete barycenter whose support size is bounded by

$$K^* \leq \sum_{i=1}^n m_i - n + 1,$$

see the LP characterization and sparsity theorem of *Anderes, Borgwardt, and Miller* (2016, Theorems 1–2)[Anderes et al., 2016]. Consequently, even though an optimal barycenter is sparse, the ambient LP may be very large because the candidate support set can grow combinatorially with $\{m_i\}$, which leads to substantial computational cost.

When computing Wasserstein barycenters, we adopt the approach of [Cuturi and Doucet, 2014], which approximates the barycenter by a discrete probability measure whose support size is moderate. The algorithm alternates between (i) updating the weights of each support point and (ii) updating the support of the barycenter. Within each iteration, entropy-regularized optimal transport is used to improve efficiency, which is a strictly convex problem. For discrete measures $\mu = \sum_p a_p \delta_{x_p}$ and $\nu = \sum_q b_q \delta_{y_q}$ with cost $M_{pq} = \|x_p - y_q\|^2$, entropy regularized OT can be formulated as

$$\min_{T \in \Pi(\mu, \nu)} \langle T, M \rangle - \varepsilon H(T), \quad H(T) = - \sum_{p,q} T_{pq} \log T_{pq},$$

whose dual involves potentials (f, g) and can be computed efficiently by Sinkhorn matrix scaling.

A.2 Wasserstein K -medians clustering

Given a set of distributions $\{\mu_i\}_{i=1}^n \subset \mathcal{P}_2(\mathbb{R}^d)$, the goal of Wasserstein K -median clustering is to find partitions $\{\mathcal{C}_k\}_{k=1}^K$ and cluster representatives $\{m_k\}_{k=1}^K$ that minimize the total within-cluster Wasserstein distance:

$$\min_{\{\mathcal{C}_k, \nu_k\}_{k=1}^K} \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \mathcal{W}_2(\mu_i, \nu_k), \quad (11)$$

where ν_k is the Wasserstein Fréchet median of the k -th cluster $\mathcal{C}_k$. This optimization problem is highly nonconvex [Le Gouic and Loubes, 2017]. We develop Algorithm 3 to approximate the solution to (11).

We elaborate two key steps in the above algorithm: initialization via PAM and the computation of Fréchet median in Wasserstein space.

To enhance stability and accelerate convergence, we use Partitioning Around Medoids (PAM) [Kaufman and Rousseeuw, 1987] as a warm start. Formally, given empirical measures $\{\mu_i\}_{i=1}^n \subset \mathcal{P}_2(X)$, we first compute the pairwise Wasserstein distance matrix

$$D_{ij} = \mathcal{W}_2(\mu_i, \mu_j), \quad i, j = 1, \dots, n. \quad (12)$$

The PAM heuristic is then applied to identify K representative medoids that minimize the total within-cluster dispersion:

$$\min_{\{\mathcal{C}_k, m_k\}} \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} D_{i, m_k}. \quad (13)$$

The resulting medoids $\{m_k^{(0)}\}_{k=1}^K$ and initial cluster assignments $\{l_i^{(0)}\}_{i=1}^n$ provide a good initialization for the subsequent Wasserstein K -median optimization.

The Fréchet median of a set of probability measures $\{\mu_i\}_{i=1}^N \subset \mathcal{P}_2(\mathbb{R}^d)$ is defined as the probability measure that minimizes the sum of its Wasserstein distances to each element in that set:

$$\mu^* = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{i=1}^N \mathcal{W}_2(\nu, \mu_i). \quad (14)$$

Algorithm 3 Wasserstein K -median Clustering with PAM Warm Start

```
1: Input: a collection of probability  $\{\mu_1, \dots, \mu_n\} \subset \mathcal{P}_2(\mathbb{R}^d)$ ; number of clusters  $K$ ; tolerance  $\varepsilon$ 
2: Output: cluster labels  $\{l_i\}_{i=1}^n$  and Wasserstein medians  $\{m_k\}_{k=1}^K$ 
3: PAM warm start:
   Compute pairwise distances  $D_{ij} = \mathcal{W}_2(\mu_i, \mu_j)$ 
   Run PAM on  $D$  to obtain initial medoids  $\mu_{n_1}, \dots, \mu_{n_K}$  and initial labels  $\{l_i^{(0)}\}_{i=1}^n$ 
   Set  $m_k^{(0)} \leftarrow \mu_{n_k}$  for  $k = 1, \dots, K$ 
4:  $t \leftarrow 0$ 
5: repeat
6:   for  $i = 1, \dots, n$  do
7:      $l_i^{(t)} \leftarrow \arg \min_{1 \leq k \leq K} \mathcal{W}_2(\mu_i, m_k^{(t)})$ 
8:   end
9:   for  $k = 1, \dots, K$  do
10:     $\mathcal{C}_k^{(t)} \leftarrow \{i : l_i^{(t)} = k\}$ 
11:     $m_k^{(t+1)} \leftarrow \arg \min_{\nu \in \mathcal{P}_2(X)} \sum_{i \in \mathcal{C}_k^{(t)}} \mathcal{W}_2(\mu_i, \nu)$ 
12:   end
13:    $\Delta^{(t)} \leftarrow \max_{1 \leq k \leq K} \mathcal{W}_2(m_k^{(t+1)}, m_k^{(t)})$ 
14:    $t \leftarrow t + 1$ 
15: until  $\Delta^{(t-1)} \leq \varepsilon$ 
16: return  $\{l_i^{(t-1)}\}_{i=1}^n$  and  $\{m_k^{(t)}\}_{k=1}^K$ 
```

To compute μ^* , we adopt an *iteratively reweighted least squares* (IRLS) scheme [You et al., 2025] defined on the Wasserstein space, which reformulates the median estimation problem as a sequence of weighted Wasserstein barycenter computations. At each iteration t , the algorithm updates the weights according to the current distances between the candidate median $\nu^{(t)}$ and the input measures:

$$w_n^{(t)} = \frac{1}{\mathcal{W}_2(\nu^{(t)}, \mu_n)}, \quad \tilde{w}_n^{(t)} = \frac{w_n^{(t)}}{\sum_{k=1}^N w_k^{(t)}}, \quad (15)$$

Given these normalized weights $\tilde{w}_n^{(t)}$, the next iterate $\nu^{(t+1)}$ is obtained by solving the corresponding weighted barycenter problem:

$$\nu^{(t+1)} = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{n=1}^N \tilde{w}_n^{(t)} \mathcal{W}_2^2(\nu, \mu_n). \quad (16)$$

Such updates replace the non smooth median objective with a locally quadratic surrogate, yielding a smooth majorization–minimization step in Wasserstein space.

In practice, each weighted barycenter subproblem (16) can be efficiently solved using the semi-free support barycenter method introduced in the previous subsection. By fixing a moderate number of support points and jointly optimizing their locations and weights under entropic regularization, we obtain a numerically stable and computationally tractable approximation of the Wasserstein median. The resulting estimator can thus be viewed as a semi-free support Wasserstein median, whose computation is summarized in Algorithm 4 [You et al., 2025].

A.3 Implementation of empirical WSD

For each pair $(\hat{\mu}_{q,m}, \hat{\mu}_{i,m})$ of empirical measures with uniform weights $1/m$, we solve the discrete optimal transport problem exactly via linear programming using the `ot.emd` routine in the Python Optimal Transport library [Flamary et al., 2021], with squared Euclidean ground cost. This yields an optimal transport plan $\pi_{q,i} \in \mathbb{R}_+^{m \times m}$ and the squared Wasserstein distance $\mathcal{W}_2^2(\hat{\mu}_{q,m}, \hat{\mu}_{i,m}) = \sum_{j,k} \pi_{q,i}(j,k) \|X_{q,j} - X_{i,k}\|^2$. We then form the barycentric projection

$$T_{\hat{\mu}_{q,m}, \hat{\mu}_{i,m}}(X_{q,j}) := m \sum_{k=1}^m X_{i,k} \pi_{q,i}(j,k),$$

Algorithm 4 Wasserstein median computation by IRLS.

```

1: Input: a collection of probability  $\{\mu_1, \dots, \mu_n\} \subset \mathcal{P}_2(\mathbb{R}^d)$ 
2: Output:  $\nu^* = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{n=1}^N \mathcal{W}_2(\nu, \mu_n)$ .
3: Initialize  $\nu^{(0)} \in \mathcal{P}_2(\mathbb{R}^d)$ .
4:
5: repeat
6:    $w_n^{(t)} = 1/\mathcal{W}_2(\nu^{(t)}, \mu_n)$ 
7:    $\tilde{w}_n^{(t)} = w_n^{(t)} / \sum_{k=1}^N w_k^{(t)}$ 
8:    $\nu^{(t+1)} = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{n=1}^N \tilde{w}_n^{(t)} \mathcal{W}_2^2(\nu, \mu_n)$ 
9:   if  $\nu^{(t+1)}$  equals one of the  $\mu_n$ 's or  $\nu^{(t+1)}$  is close to  $\nu^{(t)}$  then
10:    return  $\nu^{(t+1)}$  and stop
11: until convergence

```

which serves as the discrete analogue of the optimal transport map in Definition 2.1. When $\mathcal{W}_2(\hat{\mu}_{q,m}, \hat{\mu}_{i,m}) < \tau$ for a small tolerance τ , we set the corresponding direction vector to zero, following the convention in Definition 2.1.

B Main proofs

B.1 Proof of Theorem 3.1

Notation and preliminaries Throughout this appendix, given a metric space (Y, ρ) we write $B_\rho(y, r) := \{y' \in Y : \rho(y', y) < r\}$ for the open ball. The *Hausdorff distance* between two nonempty subsets $A, B \subseteq Y$ is

$$H(A, B) := \max \left\{ \sup_{a \in A} \inf_{b \in B} \rho(a, b), \sup_{b \in B} \inf_{a \in A} \rho(a, b) \right\},$$

which is small whenever every point of A has a close neighbor in B and vice versa, and is the natural metric in which we will compare two finite center sets. The *support* of a Borel probability measure $\mathbf{P}$ on Y , denoted $\text{supp}(\mathbf{P})$, is the set of points $y \in Y$ such that $\mathbf{P}(B_\rho(y, r)) > 0$ for every $r > 0$; on a Polish space it is nonempty and carries full mass [Billingsley, 1999].

For a finite set $A \subset \mathcal{P}_2(X)$ and $\mathbf{P} \in \mathcal{P}(\mathcal{P}_2(X))$, recall the population K -means functional and its minimum value:

$$\begin{aligned} \Phi(A, \mathbf{P}) &= \int g_A(\mu) \mathbf{P}(d\mu), \quad g_A(\mu) := \min_{\nu \in A} \mathcal{W}_2^2(\mu, \nu), \\ m_K(\mathbf{P}) &= \inf \{ \Phi(A, \mathbf{P}) : |A| \leq K \}. \end{aligned}$$

We write $\mathbf{P}_n := \frac{1}{n} \sum_{i=1}^n \delta_{\mu_i}$ for the empirical analogue of $\mathbf{P}$. To make H well-defined as a metric on center sets, we work with the parameter space

$$\mathcal{A}_K := \{A \subset \mathcal{P}_2(X) : |A| \leq K\},$$

equipped with the Hausdorff distance H induced by $\mathcal{W}_2$, and we write $D := \text{diam}(\mathcal{P}_2(X))$. With this notation, $\bar{A}(K)$ is the population K -means optimum, $A_n(K)$ is its empirical counterpart, and strong consistency means $A_n(K) \rightarrow \bar{A}(K)$ a.s. in H .

Remark 1 (Compactness of $(\mathcal{P}_2(X), \mathcal{W}_2)$). The whole argument relies on $(\mathcal{P}_2(X), \mathcal{W}_2)$ being compact, which is not obvious because $\mathcal{P}_2(X)$ is an infinite-dimensional space of measures. We therefore record the three classical facts that establish it.

(a) *Weak compactness of $\mathcal{P}(X)$* . For any $\varepsilon > 0$, since X is compact, the trivial choice $K_\varepsilon := X$ gives

$$\mu(K_\varepsilon) = \mu(X) = 1 \geq 1 - \varepsilon \quad \text{for every } \mu \in \mathcal{P}(X),$$

so the family $\mathcal{P}(X)$ is uniformly tight. Prokhorov's theorem [Prokhorov, 1956, Billingsley, 1999] then yields

$$(\mathcal{P}(X), \text{weak topology}) \text{ is compact and Polish.}$$

(b) *Equality* $\mathcal{P}_2(X) = \mathcal{P}(X)$. Fix any $x_0 \in X$. For every $\mu \in \mathcal{P}(X)$,

$$\int_X d(x, x_0)^2 \mu(dx) \leq \int_X (\text{diam}(X))^2 \mu(dx) = \text{diam}(X)^2 < \infty,$$

so the second-moment requirement defining $\mathcal{P}_2(X)$ is automatic and

$$\mathcal{P}_2(X) = \mathcal{P}(X) \quad \text{as sets.}$$

(c) $\mathcal{W}_2$ metrizes weak convergence on $\mathcal{P}_2(X)$. By Villani, 2009, Theorem 6.9 and Ambrosio et al., 2008, Proposition 7.1.5, on a compact base space

$$\mathcal{W}_2(\mu_n, \mu) \longrightarrow 0 \iff \mu_n \rightharpoonup \mu,$$

i.e., the topology induced by $\mathcal{W}_2$ coincides with the weak topology appearing in (a).

Combining (a)–(c),

$$(\mathcal{P}_2(X), \mathcal{W}_2) \text{ is a compact Polish space, and } D := \text{diam}(\mathcal{P}_2(X)) \leq \text{diam}(X) < \infty,$$

where the diameter bound follows from

$$\mathcal{W}_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^2 \pi(dx, dy) \leq \text{diam}(X)^2 \quad \text{for any } \mu, \nu \in \mathcal{P}_2(X).$$

(d) *Hausdorff compactness of $\mathcal{A}_K$* . By the Blaschke selection theorem [Burago et al., 2001, Theorem 7.3.8], for any compact metric space (Y, ρ) ,

$$(\text{Cl}(Y), H_\rho) \text{ is compact,} \quad \text{Cl}(Y) := \{F \subseteq Y : F \text{ nonempty closed}\},$$

where H_ρ is the Hausdorff distance induced by ρ . Applied to $Y = \mathcal{P}_2(X)$ and noting that

$$\mathcal{A}_K = \bigcup_{j=1}^K \{F \in \text{Cl}(\mathcal{P}_2(X)) : |F| = j\}$$

is a closed subset of $\text{Cl}(\mathcal{P}_2(X))$, we conclude

$$(\mathcal{A}_K, H) \text{ is compact.}$$

The two compactness statements

$$(\mathcal{P}_2(X), \mathcal{W}_2) \text{ compact} \quad \text{and} \quad (\mathcal{A}_K, H) \text{ compact}$$

guarantee, respectively, that empirical centers cannot escape to infinity (used in Steps 1–2) and that any sequence of empirical center sets has a Hausdorff-convergent subsequence (used in Step 3); both will be invoked repeatedly below. In addition, every g_A satisfies the uniform bound

$$0 \leq g_A(\mu) = \min_{\nu \in A} \mathcal{W}_2^2(\mu, \nu) \leq D^2 \quad \text{for all } A \in \mathcal{A}_K, \mu \in \mathcal{P}_2(X),$$

which we will use silently when invoking the SLLN on bounded random variables.

The proof adapts the classical strategy of Pollard [1981] from Euclidean to Wasserstein space, in three steps. Step 1 shows that empirical centers cannot all escape: at least one center of $A_n(K)$ stays inside a fixed ball around a chosen anchor μ_0 . Step 2 strengthens this to all centers, using the strict gap $m_{K-1}(\mathbf{P}) > m_K(\mathbf{P})$ from assumption (iv): if some center wandered far away, removing it would barely change the cost, contradicting the $K \mapsto K-1$ improvement. Step 3 then uses Hausdorff compactness of the resulting bounded family of center sets, together with uniform convergence of the empirical objective, to identify every limit point with the unique population minimizer $A^*(K)$.

The two ingredients feeding into Steps 1–3 are the following lemmas, which we record up front because Step 2 already needs the convergence $m_{K-1}(\mathbf{P}_n) \rightarrow m_{K-1}(\mathbf{P})$, itself a corollary of Lemma 2.

Lemma 1 (Lipschitz continuity of Φ). For any $A, B \in \mathcal{A}_K$ and any probability measure $\mathbf{Q}$ on $\mathcal{P}_2(X)$,

$$|\Phi(A, \mathbf{Q}) - \Phi(B, \mathbf{Q})| \leq (2D + H(A, B)) H(A, B).$$

In particular, the bound is uniform in $\mathbf{Q}$, and $A \mapsto \Phi(A, \mathbf{Q})$ is continuous on $(\mathcal{A}_K, H)$.

The point of phrasing the bound uniformly in $\mathbf{Q}$ is that we will apply it both with $\mathbf{Q} = \mathbf{P}$ and with $\mathbf{Q} = \mathbf{P}_n$; the Lipschitz constant $2D + H(A, B)$ depends only on the diameter of the ambient space, not on which measure is being integrated against.

Proof. Set $\delta := H(A, B)$. By the definition of the Hausdorff distance, for every $a \in A$ there exists $b(a) \in B$ with $\mathcal{W}_2(a, b(a)) \leq \delta$, and symmetrically. Pick any $\mu \in \mathcal{P}_2(X)$ and let $a^* \in A$ achieve $g_A(\mu) = \mathcal{W}_2^2(\mu, a^*)$. Then $b(a^*) \in B$, and the triangle inequality gives

$$\mathcal{W}_2(\mu, b(a^*)) \leq \mathcal{W}_2(\mu, a^*) + \mathcal{W}_2(a^*, b(a^*)) \leq \sqrt{g_A(\mu)} + \delta.$$

Squaring and using $\sqrt{g_A(\mu)} \leq D$ (since $g_A(\mu) \leq D^2$ on the compact ambient space),

$$g_B(\mu) \leq \mathcal{W}_2^2(\mu, b(a^*)) \leq (\sqrt{g_A(\mu)} + \delta)^2 = g_A(\mu) + 2\delta\sqrt{g_A(\mu)} + \delta^2 \leq g_A(\mu) + (2D + \delta)\delta.$$

The same argument with the roles of A and B exchanged gives $g_A(\mu) \leq g_B(\mu) + (2D + \delta)\delta$, so $|g_A(\mu) - g_B(\mu)| \leq (2D + \delta)\delta$ pointwise in μ . Integrating against $\mathbf{Q}$ and noting that the bound does not depend on $\mathbf{Q}$ proves the lemma. $\square$

Lemma 2 (Uniform SLLN over $\mathcal{A}_K$).

$$\sup_{A \in \mathcal{A}_K} |\Phi(A, \mathbf{P}_n) - \Phi(A, \mathbf{P})| \longrightarrow 0 \quad \text{a.s.}$$

In particular, $m_j(\mathbf{P}_n) \rightarrow m_j(\mathbf{P})$ a.s. for every $j \geq 1$.

For any *fixed* center set A , $\Phi(A, \mathbf{P}_n)$ is a sample average of i.i.d. random variables bounded by D^2 and converges a.s. to $\Phi(A, \mathbf{P})$ by the classical SLLN; the difficulty here is that the empirical optimizer $A_n(K)$ is itself *random*, so we will need convergence to be uniform over all admissible A . The mechanism is standard: Lemma 1 provides a uniform Lipschitz bound, and compactness of $(\mathcal{A}_K, H)$ provides a finite net, so a uniform statement reduces to a finite-maximum statement that the ordinary SLLN handles directly.

Proof. Fix $\varepsilon > 0$ and choose $\delta \in (0, 1]$ small enough that $(2D + 1)\delta < \varepsilon/3$. By compactness of $(\mathcal{A}_K, H)$ (Remark 1), there exists a finite δ -net $\mathcal{A}_{K,\delta} \subset \mathcal{A}_K$, i.e., for every $A \in \mathcal{A}_K$ there is some $A^\delta \in \mathcal{A}_{K,\delta}$ with $H(A, A^\delta) \leq \delta$. Lemma 1 then gives, for any probability measure $\mathbf{Q}$ on $\mathcal{P}_2(X)$ and any such pair (A, A^δ) ,

$$|\Phi(A, \mathbf{Q}) - \Phi(A^\delta, \mathbf{Q})| \leq (2D + \delta)\delta \leq (2D + 1)\delta < \varepsilon/3. \quad (17)$$

Each function g_{A^δ} is bounded by D^2 and the net $\mathcal{A}_{K,\delta}$ is finite, so the classical strong law of large numbers [Billingsley, 1995, Theorem 22.1], applied to each A^δ separately, yields

$$\max_{A^\delta \in \mathcal{A}_{K,\delta}} |\Phi(A^\delta, \mathbf{P}_n) - \Phi(A^\delta, \mathbf{P})| \xrightarrow{\text{a.s.}} 0.$$

For arbitrary $A \in \mathcal{A}_K$ pick the corresponding nearest $A^\delta \in \mathcal{A}_{K,\delta}$, and apply (17) once with $\mathbf{Q} = \mathbf{P}$ and once with $\mathbf{Q} = \mathbf{P}_n$. The triangle inequality, in the spirit of the empirical-process scheme of Gänssler and Stute [1979], gives

$$\begin{aligned} |\Phi(A, \mathbf{P}_n) - \Phi(A, \mathbf{P})| &\leq |\Phi(A, \mathbf{P}_n) - \Phi(A^\delta, \mathbf{P}_n)| + |\Phi(A^\delta, \mathbf{P}_n) - \Phi(A^\delta, \mathbf{P})| \\ &\quad + |\Phi(A^\delta, \mathbf{P}) - \Phi(A, \mathbf{P})|. \end{aligned}$$

The first and third terms are each at most $(2D + 1)\delta < \varepsilon/3$ by (17), while the middle term tends to 0 a.s. by the finite-maximum statement above. Taking $\sup_{A \in \mathcal{A}_K}$ on the left and then $\limsup_n$ a.s. bounds the supremum by $2(2D + 1)\delta < 2\varepsilon/3 < \varepsilon$. Letting $\varepsilon \downarrow 0$ yields the uniform convergence. The convergence $m_j(\mathbf{P}_n) \rightarrow m_j(\mathbf{P})$ then follows from $|m_j(\mathbf{P}_n) - m_j(\mathbf{P})| \leq \sup_{A \in \mathcal{A}_j} |\Phi(A, \mathbf{P}_n) - \Phi(A, \mathbf{P})| \leq \sup_{A \in \mathcal{A}_K} |\Phi(A, \mathbf{P}_n) - \Phi(A, \mathbf{P})|$, since $\mathcal{A}_j \subseteq \mathcal{A}_K$ for $j \leq K$. $\square$

Step 1: At least one empirical center lies in a fixed ball. To rule out the possibility that all centers of $A_n(K)$ drift to regions where $\mathbf{P}$ places no mass, we will fix an anchor μ_0 inside the support of $\mathbf{P}$ and show that, were every center to leave a sufficiently large ball around μ_0 , the empirical cost $\Phi(A_n(K), \mathbf{P}_n)$ would explode—contradicting the optimality of $A_n(K)$, whose cost is automatically dominated by that of the singleton competitor $\{\mu_0\}$.

Since $\mathcal{P}_2(X)$ is Polish (Remark 1), $\text{supp}(\mathbf{P})$ is nonempty; choose $\mu_0 \in \text{supp}(\mathbf{P})$. By the definition of support,

$$p_r := \mathbf{P}(B_{\mathcal{W}_2}(\mu_0, r)) > 0 \quad \text{for every } r > 0, \quad (18)$$

so any ball around μ_0 has strictly positive $\mathbf{P}$ -mass. We now fix some $r > 0$ once and for all (its specific value is irrelevant; any $r > 0$ works because of (18)), and argue by contradiction: suppose that for infinitely many n ,

$$A_n(K) \cap B_{\mathcal{W}_2}(\mu_0, M) = \emptyset. \quad (19)$$

Under (19), every center $\nu \in A_n(K)$ satisfies $\mathcal{W}_2(\nu, \mu_0) \geq M$. For any $\mu \in B_{\mathcal{W}_2}(\mu_0, r)$ and any such ν , the triangle inequality gives

$$\mathcal{W}_2(\mu, \nu) \geq \mathcal{W}_2(\nu, \mu_0) - \mathcal{W}_2(\mu, \mu_0) \geq M - r.$$

Therefore $g_{A_n(K)}(\mu) = \min_{\nu} \mathcal{W}_2^2(\mu, \nu) \geq (M - r)^2$ for every $\mu \in B_{\mathcal{W}_2}(\mu_0, r)$. Restricting the integral defining Φ to this ball yields the lower bound

$$\begin{aligned} \Phi(A_n(K), \mathbf{P}_n) &= \int g_{A_n(K)}(\mu) \mathbf{P}_n(d\mu) \\ &\geq \int_{B_{\mathcal{W}_2}(\mu_0, r)} (M - r)^2 \mathbf{P}_n(d\mu) = (M - r)^2 \mathbf{P}_n(B_{\mathcal{W}_2}(\mu_0, r)). \end{aligned} \quad (20)$$

The function $\mathbf{1}_{B_{\mathcal{W}_2}(\mu_0, r)}$ is bounded, so the SLLN gives $\mathbf{P}_n(B_{\mathcal{W}_2}(\mu_0, r)) \rightarrow p_r$ a.s. Taking $\liminf$ in (20),

$$\liminf_{n \rightarrow \infty} \Phi(A_n(K), \mathbf{P}_n) \geq (M - r)^2 p_r. \quad (21)$$

To produce a contradiction we now derive a matching upper bound. The singleton $\{\mu_0\}$ has cardinality $1 \leq K$ and is therefore admissible, so by optimality of $A_n(K)$,

$$\Phi(A_n(K), \mathbf{P}_n) \leq \Phi(\{\mu_0\}, \mathbf{P}_n) = \int \mathcal{W}_2^2(\mu, \mu_0) \mathbf{P}_n(d\mu).$$

The integrand is bounded by D^2 , so the SLLN gives $\Phi(\{\mu_0\}, \mathbf{P}_n) \rightarrow \Phi(\{\mu_0\}, \mathbf{P}) =: c_0$ a.s., where $c_0 = \int \mathcal{W}_2^2(\mu, \mu_0) \mathbf{P}(d\mu) \leq D^2 < \infty$. Hence

$$\limsup_{n \rightarrow \infty} \Phi(A_n(K), \mathbf{P}_n) \leq c_0. \quad (22)$$

Combining (21) and (22),

$$(M - r)^2 p_r \leq c_0.$$

The right-hand side c_0 is a fixed finite constant independent of M , while the left-hand side grows without bound as $M \rightarrow \infty$. Therefore choosing $M = M_1$ large enough that $(M_1 - r)^2 p_r > c_0$ contradicts (19). We conclude that, almost surely, $A_n(K) \cap B_{\mathcal{W}_2}(\mu_0, M_1) \neq \emptyset$ for all n large enough.

Step 2: All empirical centers lie in a larger ball. Step 1 controls only one center; the others could in principle still drift away. To rule this out we use the strict gap $m_{K-1}(\mathbf{P}) > m_K(\mathbf{P})$ from assumption (iv). The strategy is to argue that if some center ν_n wandered very far from $\mathbf{P}$'s mass, then on the bulk of the data ν_n would never be the closest center and could therefore be dropped from $A_n(K)$ without significantly raising the empirical cost. The resulting reduced set, of size only $K - 1$, would then provide a near-optimal solution at level $K - 1$, contradicting the strict $K \mapsto K - 1$ improvement guaranteed by (iv).

Let M_1 be the constant from Step 1, and suppose, for contradiction, that along a subsequence (still indexed by n) there exists $\nu_n \in A_n(K)$ with

$$\mathcal{W}_2(\nu_n, \mu_0) > 5M_1. \quad (23)$$

Step 1 also guarantees the existence of an in-ball center $\eta_n \in A_n(K)$ with $\mathcal{W}_2(\eta_n, \mu_0) \leq M_1$. Define the reduced set

$$A'_n := A_n(K) \setminus \{\nu_n\}, \quad |A'_n| = K - 1.$$

Since $A'_n \subset A_n(K)$, the increment $g_{A'_n}(\mu) - g_{A_n(K)}(\mu)$ is nonnegative for every μ . We now bound it pointwise, distinguishing whether μ lies in the inner ball $B_{\mathcal{W}_2}(\mu_0, 2M_1)$ or in its complement.

The factor 5 in (23) is chosen precisely so that on the inner ball, η_n is always strictly closer to μ than ν_n is: for every μ with $\mathcal{W}_2(\mu, \mu_0) \leq 2M_1$, the triangle inequality gives

$$\mathcal{W}_2(\mu, \nu_n) \geq \mathcal{W}_2(\nu_n, \mu_0) - \mathcal{W}_2(\mu, \mu_0) > 5M_1 - 2M_1 = 3M_1,$$

$$\mathcal{W}_2(\mu, \eta_n) \leq \mathcal{W}_2(\mu, \mu_0) + \mathcal{W}_2(\mu_0, \eta_n) \leq 2M_1 + M_1 = 3M_1.$$

Hence $\mathcal{W}_2(\mu, \eta_n) \leq \mathcal{W}_2(\mu, \nu_n)$, so the minimum $\min_{\nu \in A_n(K)} \mathcal{W}_2(\mu, \nu)$ is attained at some center other than ν_n (one example being η_n). Removing ν_n therefore does not change the minimum on the inner ball:

$$g_{A'_n}(\mu) = g_{A_n(K)}(\mu) \quad \text{for every } \mu \text{ with } \mathcal{W}_2(\mu, \mu_0) \leq 2M_1. \quad (24)$$

On the outer region $\mathcal{W}_2(\mu, \mu_0) > 2M_1$, the in-ball center $\eta_n \in A'_n$ provides an explicit upper estimate. Using $g_{A_n(K)}(\mu) \geq 0$ and the triangle inequality,

$$\begin{aligned} g_{A'_n}(\mu) - g_{A_n(K)}(\mu) &\leq g_{A'_n}(\mu) \leq \mathcal{W}_2^2(\mu, \eta_n) \\ &\leq (\mathcal{W}_2(\mu, \mu_0) + \mathcal{W}_2(\mu_0, \eta_n))^2 \leq (\mathcal{W}_2(\mu, \mu_0) + M_1)^2. \end{aligned}$$

In this region $M_1 < \frac{1}{2}\mathcal{W}_2(\mu, \mu_0)$, so $(\mathcal{W}_2(\mu, \mu_0) + M_1)^2 < (\frac{3}{2}\mathcal{W}_2(\mu, \mu_0))^2 = \frac{9}{4}\mathcal{W}_2^2(\mu, \mu_0)$, which together with (24) gives the pointwise bound

$$0 \leq g_{A'_n}(\mu) - g_{A_n(K)}(\mu) \leq \begin{cases} 0, & \mathcal{W}_2(\mu, \mu_0) \leq 2M_1, \\ \frac{9}{4}\mathcal{W}_2^2(\mu, \mu_0), & \mathcal{W}_2(\mu, \mu_0) > 2M_1. \end{cases} \quad (25)$$

We now integrate (25) against $\mathbf{P}_n$. Splitting the domain into the inner ball (where the increment vanishes) and the outer region,

$$\begin{aligned} \Phi(A'_n, \mathbf{P}_n) - \Phi(A_n(K), \mathbf{P}_n) &= \int [g_{A'_n}(\mu) - g_{A_n(K)}(\mu)] \mathbf{P}_n(d\mu) \\ &\leq \frac{9}{4} \int_{\{\mathcal{W}_2(\mu, \mu_0) > 2M_1\}} \mathcal{W}_2^2(\mu, \mu_0) \mathbf{P}_n(d\mu). \end{aligned} \quad (26)$$

The integrand on the right is bounded by D^2 , so the SLLN gives a.s.

$$\frac{9}{4} \int_{\{\mathcal{W}_2(\mu, \mu_0) > 2M_1\}} \mathcal{W}_2^2(\mu, \mu_0) \mathbf{P}_n(d\mu) \xrightarrow{n \rightarrow \infty} \frac{9}{4} \int_{\{\mathcal{W}_2(\mu, \mu_0) > 2M_1\}} \mathcal{W}_2^2(\mu, \mu_0) \mathbf{P}(d\mu) =: \tau(M_1). \quad (27)$$

The function $\mu \mapsto \mathcal{W}_2^2(\mu, \mu_0)$ is integrable against $\mathbf{P}$ (it is bounded by D^2), and the indicator $\mathbf{1}_{\{\mathcal{W}_2(\mu, \mu_0) > 2M_1\}}$ converges $\mathbf{P}$ -a.e. to 0 as $M_1 \rightarrow \infty$. Dominated convergence therefore gives $\tau(M_1) \rightarrow 0$ as $M_1 \rightarrow \infty$, so the right-hand side of (26) can be made arbitrarily small by choosing M_1 large.

For the lower bound on the same difference, we use that $|A'_n| = K - 1$ and $|A_n(K)| = K$, and the optimality of the empirical K - and $(K - 1)$ -means:

$$\Phi(A'_n, \mathbf{P}_n) \geq m_{K-1}(\mathbf{P}_n), \quad \Phi(A_n(K), \mathbf{P}_n) = m_K(\mathbf{P}_n).$$

Lemma 2 applied at levels $K - 1$ and K gives $m_{K-1}(\mathbf{P}_n) \rightarrow m_{K-1}(\mathbf{P})$ and $m_K(\mathbf{P}_n) \rightarrow m_K(\mathbf{P})$ a.s., so

$$\liminf_{n \rightarrow \infty} [\Phi(A'_n, \mathbf{P}_n) - \Phi(A_n(K), \mathbf{P}_n)] \geq m_{K-1}(\mathbf{P}) - m_K(\mathbf{P}) =: \Delta_K > 0, \quad (28)$$

where strict positivity of Δ_K is exactly assumption (iv).

The two estimates (26)–(27) and (28) pull in opposite directions: the former says the difference $\Phi(A'_n, \mathbf{P}_n) - \Phi(A_n(K), \mathbf{P}_n)$ is eventually at most $\tau(M_1) + o(1)$ a.s., while the latter says it is eventually at least $\Delta_K - o(1)$ a.s. Choosing M_1 large enough that $\tau(M_1) < \Delta_K/2$ yields, for n large, $\Delta_K - o(1) \leq \Phi(A'_n, \mathbf{P}_n) - \Phi(A_n(K), \mathbf{P}_n) \leq \tau(M_1) + o(1) < \Delta_K/2 + o(1)$, which is impossible. Hence the assumption (23) of a runaway center fails for M_1 large; equivalently, almost surely

$$A_n(K) \subseteq B_{\mathcal{W}_2}(\mu_0, 5M_1) \quad \text{for all } n \text{ large enough.}$$

Step 3: Identification of the limit. Steps 1 and 2 confine $A_n(K)$ to a fixed compact ball, so by Hausdorff compactness it has limit points; the remaining task is to show that any such limit point coincides with the population minimizer $\bar{A}(K)$. The two ingredients are Lemma 2, which lets us pass from $\Phi(A_n(K), \mathbf{P}_n)$ to $\Phi(A_n(K), \mathbf{P})$, and Lemma 1, which lets us pass from $\Phi(A_n(K), \mathbf{P})$ to $\Phi(A^*, \mathbf{P})$ along a Hausdorff-convergent subsequence. Together they identify the population value of any limit point A^* as $m_K(\mathbf{P})$, and uniqueness assumption (iii) then forces $A^* = \bar{A}(K)$.

By Step 2, almost surely $A_n(K)$ eventually lies in the closed ball $\bar{B} := \overline{B_{\mathcal{W}_2}(\mu_0, 5M_1)}$, which is compact since $(\mathcal{P}_2(X), \mathcal{W}_2)$ is compact (Remark 1). Therefore

$$\mathcal{A}_K^{\bar{B}} := \{A \subset \bar{B} : |A| \leq K\}$$

is compact under the Hausdorff distance (again by Blaschke selection). Hence every subsequence of $\{A_n(K)\}$ admits a Hausdorff-convergent further subsequence: passing to such a subsequence (still indexed by j), we have $A_{n_j}(K) \rightarrow A^*$ in H for some $A^* \in \mathcal{A}_K^{\bar{B}}$.

We compute $\Phi(A^*, \mathbf{P})$ along this subsequence in two steps. First, by Lemma 1, $\Phi(\cdot, \mathbf{P})$ is continuous in H , so

$$\Phi(A_{n_j}(K), \mathbf{P}) \rightarrow \Phi(A^*, \mathbf{P}). \quad (29)$$

Second, by Lemma 2, $\sup_A |\Phi(A, \mathbf{P}_{n_j}) - \Phi(A, \mathbf{P})| \rightarrow 0$ a.s., and applying this uniform statement at $A = A_{n_j}(K)$ in particular gives

$$|\Phi(A_{n_j}(K), \mathbf{P}_{n_j}) - \Phi(A_{n_j}(K), \mathbf{P})| \rightarrow 0 \quad \text{a.s.} \quad (30)$$

Combining (29) and (30) with the empirical optimality $\Phi(A_{n_j}(K), \mathbf{P}_{n_j}) = m_K(\mathbf{P}_{n_j})$ and the convergence $m_K(\mathbf{P}_{n_j}) \rightarrow m_K(\mathbf{P})$ from Lemma 2,

$$\Phi(A^*, \mathbf{P}) = \lim_j \Phi(A_{n_j}(K), \mathbf{P}) = \lim_j \Phi(A_{n_j}(K), \mathbf{P}_{n_j}) = \lim_j m_K(\mathbf{P}_{n_j}) = m_K(\mathbf{P}).$$

Thus A^* achieves the population infimum, and assumption (iii)—that $\bar{A}(K)$ is unique up to permutation—forces $A^* = \bar{A}(K)$.

We have shown that every Hausdorff-convergent subsequence of $\{A_n(K)\}$ converges to the same limit $\bar{A}(K)$. A standard subsequence-of-subsequence argument then yields convergence of the full sequence: if $A_n(K)$ did not converge to $\bar{A}(K)$, there would exist $\varepsilon > 0$ and a subsequence with $H(A_{n_j}(K), \bar{A}(K)) \geq \varepsilon$ for all j , contradicting the existence of a further sub-subsequence converging to $\bar{A}(K)$. Therefore

$$A_n(K) \rightarrow \bar{A}(K) \quad \text{a.s. in the Hausdorff metric,}$$

and the convergence $\Phi(A_n(K), \mathbf{P}_n) = m_K(\mathbf{P}_n) \rightarrow m_K(\mathbf{P})$ has already been established in Lemma 2. $\square$

B.2 Proof of Theorem 3.2

We prove the consistency of empirical RWS under any Wasserstein K -center clustering method that produces strongly consistent cluster centers. The argument combines three ingredients: (i) the assumed Hausdorff convergence of empirical cluster centers; (ii) consistency of the empirical WSD plug-in estimator under two-stage sampling (Bachoc et al. [2024, Theorem 6.3]); and (iii) the continuous mapping theorem.

Step 1: Stability of cluster assignments. Let $\{A_k^*\}_{k=1}^K$ denote the unique population centers and $\{\hat{A}_{n,k}\}_{k=1}^K$ their empirical counterparts, with $\hat{A}_{n,k} \rightarrow A_k^*$ almost surely in the Hausdorff metric induced by $\mathcal{W}_2$ by hypothesis. For any $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ that has a unique nearest population center $A_{k(\mu)}^*$, the continuity of $A \mapsto \mathcal{W}_2(\mu, A)$ on the space of finite center sets under Hausdorff distance implies that, almost surely for n large enough, μ is assigned to the empirical cluster $\hat{\mathcal{C}}_{k(\mu)}^{(n)}$ corresponding to $\hat{A}_{n,k(\mu)}$. By the well-separation assumption, this holds for $\mathbf{P}$ -almost every μ .

Step 2: Convergence of empirical cluster distributions. Let $\hat{\mathbf{P}}_{n,m}^{(k)} := \frac{1}{|\hat{\mathcal{C}}_k^{(n)}|} \sum_{\hat{\mu}_{i,m} \in \hat{\mathcal{C}}_k^{(n)}} \delta_{\hat{\mu}_{i,m}}$ be the empirical distribution over distributions in cluster k , and $\mathbf{P}^{(k)}$ its population counterpart restricted to $\mathcal{C}_k$. By Step 1 and Bachoc et al. [2024, Lemma 6.2], $\hat{\mathbf{P}}_{n,m}^{(k)} \rightarrow \mathbf{P}^{(k)}$ in $d_{\text{BL}(2)}$ in probability as $n, m \rightarrow \infty$, for each $k = 1, \dots, K$.

Step 3: Pointwise consistency of empirical WSD on each cluster. Fix q and let $k_q := k(\mu_q)$ denote the cluster index of μ_q . By Bachoc et al. [2024, Theorem 6.3] applied to the cluster-restricted distribution-of-distributions $\mathbf{P}^{(k)}$, for each $k = 1, \dots, K$,

$$\widehat{\text{WSD}}(\hat{\mu}_{q,m}; \hat{\mathbf{P}}_{n,m}^{(k)}) \xrightarrow{P} \text{WSD}(\mu_q; \mathbf{P}^{(k)}) \quad \text{as } n, m \rightarrow \infty,$$

provided $\mathbf{P}^{(k)}$ is atomless, which is inherited from the atomlessness of $\mathbf{P}$.

Step 4: Continuous mapping. By Steps 1–3 and the continuous mapping theorem applied to the function $(a, b_1, \dots, b_{K-1}) \mapsto a - \max_j b_j$,

$$\widehat{\text{RWSD}}_K(\hat{\mu}_{q,m}) = \widehat{\text{WSD}}(\hat{\mu}_{q,m}; \hat{\mathbf{P}}_{n,m}^{(k_q)}) - \max_{j \neq k_q} \widehat{\text{WSD}}(\hat{\mu}_{q,m}; \hat{\mathbf{P}}_{n,m}^{(j)}) \xrightarrow{P} \text{RWSD}_K(\mu_q),$$

which completes the proof. $\square$

B.3 Proof of Corollary 1

By Theorem 3.1, the Wasserstein K -means centers $A_{n,K}^*$ converge almost surely in the Hausdorff metric to A_K^* , which is unique up to permutation by assumption (iii). The hypotheses of Theorem 3.2 are therefore satisfied with $\hat{A}_{n,k}$ taken to be the empirical K -means centers, and the conclusion follows. $\square$

C Additional details in real data analysis

This section collects supplementary details for the real data analysis in Section 5, including an illustration of the MNIST distributional embedding and the full preprocessing pipeline for the AML flow cytometry cohort.

C.1 Illustration of MNIST distributional embedding

Each grayscale MNIST handwritten digit image of size 28×28 is treated as a discrete empirical measure over pixel coordinates with intensities normalized to sum to one,

$$\mu_i = \sum_{p=1}^{784} w_{ip} \delta_{x_p}, \quad w_{ip} = \frac{I_{ip}}{\sum_{q=1}^{784} I_{iq}},$$

where $x_p \in [0, 1]^2$ is the normalized pixel position and I_{ip} the grayscale intensity at pixel p in image i . This transforms each image into a probability distribution $\mu_i \in \mathcal{P}_2([0, 1]^2)$.

C.2 AML cytometry preprocessing details

We provide the full preprocessing pipeline for the AML cytometry experiment in Section 5.2. The P1 panel of eight markers (CD7, CD56, CD34, CD117, CD33, HLA-DR, CD13, CD45) are included in our analysis, which capture progenitor signatures, myeloid lineage, and AML-aberrant antigen expression. Each BLAST110 sample contains hundreds of thousands of single-cell flow cytometry profiles. Fluorescence intensities were first transformed using the hyperbolic arcsine with cofactor $c = 5$,

$$x^{\text{trans}} = \text{arcsinh}(x/c),$$

followed by per-marker z -score normalization computed across the pooled cells of all samples. This stabilizes variance across fluorescence channels and ensures comparability among samples. To embed each sample into Wasserstein space, we represented its single-cell data as an empirical distribution,

$$\mu_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \delta_{x_j}.$$

To make the computations feasible, we randomly selected $m = 1000$ cells from each sample without replacement and constructed the empirical distribution

$$\mu_i^{(m)} = \frac{1}{m} \sum_{j=1}^m \delta_{x_j^{(i)}}.$$

Subsampling at this scale has been shown to preserve major phenotypic structures in high-dimensional cytometry data while greatly reducing the computational cost of OT step [Samusik et al., 2016, Levine et al., 2015].

D Choices of comparison

We compare RWSD with the silhouette score and the Davies and Bouldin index because both are standard internal validation criteria and can be directly computed from the Wasserstein clustering results or pairwise Wasserstein distance matrix. This facilitates fair comparisons because the three indices are computed from the same Wasserstein clustering results. Other criteria are either computationally expensive in Wasserstein space, or not directly comparable to our method. The gap statistic requires a reference null distribution, which is difficult to choose in Wasserstein space. Stability based criteria (such as prediction strength) require repeated sample splitting, re-clustering, and recomputation of OT distances, which is prohibitive for large set of distributions. We therefore focus on baselines that can be implemented without introducing additional parameters or complicated clustering pipeline.

E Implementation Details

E.1 Computational complexity

We summarize the complexity of the main building blocks in our method. Throughout, n denotes the number of empirical distributions to be clustered, each supported on m points in $\mathbb{R}^d$, K the number of clusters, and $|\mathcal{K}|$ the number of candidate K values evaluated in Algorithm 1. We use two types of OT computations in our implementation. Exact OT maps are computed via `ot.emd` in the Python Optimal Transport library [Flamary et al., 2021] to evaluate the empirical WSD and RWSD. We use entropic regularized OT or Sinkhorn divergence to compute the pairwise validation distances, PAM warm start, DBI computation, and free support barycenter or median for computational efficiency [Cuturi, 2013, Cuturi and Doucet, 2014].

Exact discrete OT for empirical WSD and RWSD. Let $C_{\text{EMD}}(m)$ denote the computational cost of solving an exact discrete optimal transport problem between two empirical measures with m support points. For each pair $(\hat{\mu}_{q,m}, \hat{\mu}_{i,m})$, empirical WSD first solves the exact discrete OT problem using `ot.emd`, obtaining an optimal transport plan $\pi_{q,i}$. Given this plan, forming the barycentric projection $T_{\hat{\mu}_{q,m}, \hat{\mu}_{i,m}}$ costs $\mathcal{O}(m^2 d)$ per pair. Therefore, computing all pairwise transport directions needed for WSD over N empirical distributions has worst case cost

$$\mathcal{O}(n^2 C_{\text{EMD}}(m) + n^2 m^2 d).$$

Entropic OT and Sinkhorn divergence for validation distances. Let $C_{\text{Sink}}(m, \varepsilon)$ denote the computational cost of computing an entropic OT value between two m point empirical measures at regularization level ε . In the complexity bounds of Dvurechensky et al. [2018], one may take

$$C_{\text{Sink}}(m, \varepsilon) = \mathcal{O}\left(\frac{m^2}{\varepsilon^2}\right),$$

up to accuracy dependent constants. The pairwise distance matrix used by silhouette, MDS visualization, and PAM warm start is based on the square root of the Sinkhorn divergence, which requires three entropic OT computations for each pair. Thus, constructing this distance matrix for all pairs costs

$$\mathcal{O}(n^2 C_{\text{Sink}}(m, \varepsilon)).$$

This matrix is computed once and reused across all candidate values of K whenever the same dataset and regularization parameter are used.

Empirical RWSD for a fixed clustering. For a fixed partition with K clusters, RWSD computes the WSD of each empirical distribution with respect to its own cluster and with respect to other clusters. In the worst case, this requires exact OT computations over all ordered pairs of empirical distributions, giving

$$\mathcal{O}(n^2 C_{\text{EMD}}(m) + n^2 m^2 d).$$

Table 7: Average time of execution and memory required for implementation. Runtime is reported for parallel computation on 24 CPUs.

	simulation	real data analysis
runtime (per experiment)	1 hour	4 hours
memory (per CPU)	5 GB	5 GB

The final comparison between within cluster depth and between cluster depth adds only $\mathcal{O}(nK)$ operations.

Wasserstein K center clustering. The semi-free support Wasserstein K median implementation keeps a moderate support size s , typically $s \ll m$. Each assignment step computes entropic OT costs between n empirical distributions and K current centers, which costs

$$\mathcal{O}(nK C_{\text{Sink}}(s, m, \varepsilon)) ,$$

where $C_{\text{Sink}}(s, m, \varepsilon)$ denotes the Sinkhorn cost between an s point center (distribution) and an m point empirical distribution. Each center update is computed by a free support entropic Wasserstein barycenter step [Cuturi and Doucet, 2014], followed by an IRLS refinement for the Wasserstein median. Letting T_{bar} and T_{IRLS} denote the numbers of barycenter inner iterations and IRLS outer iterations, and using $\sum_{k=1}^K |C_k| = n$, the per-iteration update cost takes the form

$$\text{Update}_{K\text{-median}} = \mathcal{O}(T_{\text{IRLS}} T_{\text{bar}} n C_{\text{Sink}}(s, m, \varepsilon))$$

with the Wasserstein K -means update corresponding to $T_{\text{IRLS}} = 1$. The PAM warm start, when used, reuses the precomputed Sinkhorn distance matrix and adds $\mathcal{O}(n^2 K)$ operations for the greedy medoid search.

DBI and silhouette. The silhouette score is computed from the precomputed pairwise Sinkhorn distance matrix. The DBI computation uses the free support medians as cluster prototypes. Its within cluster scatter costs

$$\mathcal{O}(nC_{\text{Sink}}(s, m, \varepsilon)) ,$$

and its between prototype distances cost

$$\mathcal{O}(K^2 C_{\text{Sink}}(s, s, \varepsilon)) .$$

Total cost of Algorithm 1. Let T be the average number of clustering iterations until convergence. A worst case description of the total cost over all candidate values in $\mathcal{K}$ is

$$\begin{aligned} & \underbrace{\mathcal{O}(n^2 C_{\text{Sink}}(m, \varepsilon))}_{\text{pairwise Sinkhorn distances}} \\ & + |\mathcal{K}| T \underbrace{\mathcal{O}(nK C_{\text{Sink}}(s, m, \varepsilon) + \text{Update}_{K\text{-median}})}_{\text{clustering with free support representatives}} \\ & + |\mathcal{K}| \underbrace{\mathcal{O}(n^2 C_{\text{EMD}}(m) + n^2 m^2 d)}_{\text{exact OT and barycentric projections for RWSD}} . \end{aligned}$$

In our implementations, the most computationally intensive steps are the computation of pairwise Wasserstein distances, Wasserstein barycenter updates, and Wasserstein median updates. We improve the computational efficiency by restricting moderate support sizes for barycenter or Wasserstein medians, using entropic regularized optimal transport, and the iteratively reweighted least squares procedure described in Appendix A.2. The computational cost increases with the number of empirical distributions, the support size of each distribution, the candidate range of K . The time of execution and memory required for implementation are summarized in Table 7. Since we employed parallel computing in conducting simulations, the total runtime for each case is equal to the runtime per CPU.

E.2 Computational resources

All simulations and real data analysis in this paper can be reproduced on CPUs only, and no GPU or TPU acceleration is required.

These simulations can be run on a common laptop (*e.g.* Macbook Air). For reproducing the full set of synthetic experiments, we recommend a machine with 10 CPUs and at least 16 GB of memory.

The MNIST experiments represent each 28×28 grayscale image as a discrete probability distribution over pixel coordinates. Since each empirical distribution has support size at most 784, the MNIST experiments can also be reproduced on CPU only. We recommend at least 16 GB of memory for reproducing the MNIST clustering and validation results.

The AML flow cytometry experiment uses only the BLAST110 cohort from the open flow cytometry dataset of Mocking et al. [2024]. The raw BLAST110 archive is approximately 14.9 GB. We recommend at least 25 GB of free disk space for downloading and unpacking the raw data. We also recommend at least 64 GB of memory if the raw FCS files are loaded or preprocessed in large batches.

E.3 Software and data licenses

The implementation uses standard open source scientific computing packages. The analysis in this paper are conducted using Python. No proprietary software is required to reproduce the reported experiments. Table 8 summarizes the existing datasets and software used in our experiments, together with the corresponding license or access terms.

Table 8: Existing datasets and software used in the experiments.

Asset	Use in this paper	License or terms
Synthetic data	Generated by the authors for controlled simulation studies.	No external data license applies.
MNIST handwritten digit database	Used as a real distribution-valued image dataset by normalizing each image into a discrete probability distribution over pixel coordinates.	Publicly available benchmark dataset used for research evaluation. We credit the original dataset creators and follow the dataset access terms.
BLAST110 AML flow cytometry cohort	Used as the real flow cytometry dataset for the AML versus healthy control experiment in Section 5.2. We use only BLAST110 and do not use LAIP29.	Open dataset associated with Mocking et al. [2024]. The Zenodo record releases the data under the Creative Commons Attribution 4.0 International license.
NumPy and SciPy	Numerical linear algebra, array operations, and scientific computing utilities.	BSD license.
scikit-learn	Baseline utilities and clustering related routines where applicable.	BSD 3-Clause license.
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